

COMPARATIVE PHYTOCHEMICAL, ANTIOXIDANT, ANTIMICROBIAL AND ANTI INFLAMMATORY PROFILING OF SELECTED *SWERTIA* SPECIES OF UTTARAKHAND HIMALAYA

1. Introduction:

It's said that 'plant is equal to life' and rightly so as they are the most important part of the ecosystem, as they provide resources to all living creatures infinitely. Traditionally used medicinal plants are well known to the human race but their usage in the modern times has been harvested from their native habitat and handled poorly. Ancient texts are stuffed with the use of medicinal plants and their products, and they have been availed by many communities for ages to cure various diseases. Over the past three to four decades the usage of the medicinal plants has tremendously increased. The idea of natural products and medicines saw an exponential growth in the last two years all over the world after the outbreak of one of the deadliest pandemic (COVID - 19) in 2019 in order to build immunity against diseases. According to World Health Organization 2008, medicinal plants can be defined as plants that contain properties or compounds that can be used for therapeutic purpose, those that synthesize metabolites to produce useful drugs or exert beneficial pharmacological effects on the human or animal body.

According to WHO around 80% of the population of the developing countries depends on traditional medicines which mostly include plant drugs. At present, it has been estimated that, number of higher plants that are being used for medicinal purpose is more than 50,000; this number adds up to 20% of the world flora (Schippmann *et al.*, 2002). With the advancement of technology and research their demand has increased tremendously. India being rich in medicinal plants diversity has attained a defined place in world's herbal market. She is well known as the 'Medicinal Garden of the world', as around 15,000 - 20,000, out of 45,000 different plant species, have therapeutic potential (Anonymous, 2000). These herbal drugs being reasonably priced and with less or no side effects has become a better source for cure. According to Newman *et al.*, 2000 for last 30 years nearly 50% approved drugs in the field of cancer are extracted either directly or indirectly from plants. Random screening and ethno botany are the two main strategies for selecting medicinal plant species. And it is quite possible that India can create boon in the medicinal plant market if the access to good quality drug extraction from these medicinal plants become more feasible for availability.

According to Ayurvedic litterateurs *Swertia* is one of the most diverse genera which comprises of 170 known species and belongs to the Gentianaceae family, play a vital role in traditional medicine. Genus *Swertia* is widely distributed throughout the world however in India; mostly are found at high altitude of temperate Himalayas from Kashmir to Bhutan and some in the Khasia and Western Ghats hills. *Swertia* has been used globally in traditional medicines since ages. Some of the varieties are used as crude drugs in the Indian pharmacopoeia, however, traditionally in India, *Swertia* being used as a bitter tonic in stimulating appetite and as a febrifuge. Chinese traditional medicine has been using this genus for the treatment of hepatic, choleric, and inflammatory diseases. Other counties like Bhutan, Pakistan, Japan, Nepal etc. have also been using this genus in their traditional medicine.

Swertia exhibit a broad range of biological and pharmacological activities, including anticancer, antioxidant, anti-diabetic, antibacterial, anti-hepatitis, anthelmintic, anti-leishmanial, neuroprotective. *Swertia chirata*, the most used herb of this genus, is described as bitter in taste and has various pharmacological properties like anti - helmintic, hypo - glycemic and anti-pyretic. *S. chirata* is used as hepatoprotective, antiinflammatory, anti - malarial as well as anti-cancerous and reduction of hyper- proliferation (Reen *et al.*, 2001; Saha *et al.*, 2004; Saha *et al.*, 2006). Properties like anti-bacterial, anti-fungal have also been reported by various authors (Ravishankar *et al.*, 2007; Mallilarjun *et al.*, 2010). The drug obtained from the dried plant parts is used in various combinations and forms for the treatment of chronic fever, bronchial asthma, malaria, diabetes, digestive disorders etc. Significant activity against acute, sub-acute, and chronic models of inflammation has been observed for the benzene extract of *S. chirata* (whole plant). This herb is used to stimulate digestive fire (hyper acidity) and improves digestion. Regular use of this herb is known to increase the production of red blood cells playing a pivotal role in anemic patients.

The plants of the *Swertia* genus are rich sources of xanthonenes, flavonoids, irridoid and seco-irridoid glycosides, terpenoids, and alkaloids. *Swertia* species are of high pharmacological interest. Wide level applications of genus *Swertia* indicate presence of many bioactive molecules in the plant which enhances its utility for extraction of drugs. The phytochemical investigation of the genus *Swertia*, as carried out so far, has afforded numerous compounds with varying structural patterns. The chemical constituents, estimates more than twenty polyhydroxylated xanthonenes and some of these are swertinin, swerchirin, mangiferin, decussatin

and isobellidifolin; a dimeric xanthone and chiratanin has also been segregated (Tabasuum *et al.*, 2012). Swerchirin and Swertiamarin are the two major active components which have been accounted for the above properties of this herb (Saxena *et al.*, 1991). *Swertia* plants are rich sources of a variety of organic compounds of varying structural patterns, and, due to their natural distribution, are, thus, highly relevant not only for medicinal but also for chemotaxonomic studies. These herbs promote digestion, particularly of fats and also help in regulation of sugar level in blood.

2. Plan of Work

Considering the above facts, the present study will be carried out to study the species of the genus *Swertia* in the state Uttarakhand and evaluate the phytochemical activity, antioxidant potential, anti- microbial and anti-inflammatory activities of selected species of *Swertia* collected from different regions of Uttarakhand.

Aims and objectives

Present study is aimed at investigating the phytochemical, antioxidant, antimicrobial and antiinflammatory activity of *Swertia* spp. of Uttarakhand Himalaya.

Following objects are taken for the study:-

1. Collection, identification and authentication of different species of the *Swertia* found in Garhwal Himalaya.
2. Phytochemical Screening of different species of the *Swertia* found in Garhwal Himalaya.
3. Investigation of antioxidant activity of selected species of the *Swertia* found in Garhwal Himalaya.
4. Investigation of antimicrobial activity of selected species of the *Swertia* found in Garhwal Himalaya.
5. Investigation of antiinflammatory activity of selected species of the *Swertia* found in Garhwal Himalaya.

3. Review of Literature

Plants contribute towards meeting the basic requirements for human survival. They are the major source of drugs in almost every system of medicine throughout the world. By combining the botanical, ethnobotanical, phytochemical and biological techniques, a multi disciplinary approach is developed to discover a drug from a plant. One such genus with wide variety of medicinal uses is *Swertia*. It is one of the most important herbal plants which have been used since centuries for curing various disorders all over the world. It is stated as a potential medicinal herb in Unani, Ayurveda and Siddha system of medicines. Consequently, in modern medicine as well, *Swertia* is present in many drugs that are used in treatment of various disorders such as diabetes, cancer, gastro intestinal disorder, inflammations etc.

Swertia is one of the most diverse genera of medicinally important herbs which are distributed throughout the world except South America and Australia. The genus *Swertia*, belongs to family Gentianaceae which consist roughly of about 170 species all over the world (Brahmachari *et al.*, 2004). Although this genus is cosmopolitan, yet it's diversely found in temperate, sub-tropical regions and mountane tropic regions of the world. This genus is supposed to be the native of northern hemisphere of Asia while Europe, America, and Africa were the secondary regions where it originated (Kaur *et al.*, 2019). The 170 species are commonly found in the 14 regions all over the world and among all the continents, Eastern Asia ranks first in the distribution and diversity of this genus. It comprises of 86 Species of *Swertia*, out of which 58 species are indigenous. Almost 20 – 25 species of *Swertia* has been reported by in the African continent. The genus *Swertia* is represented by *S. perennis* in the western countries of Europe (Germany, Poland, Switzerland etc) (Urbaniak, 2018). While in United States of America, the genus is distributed mainly in the North – Western mountains which include states like Washington.

In Asia, this taxon is mostly found in high mountains in the altitudinal range of 1200 - 3600 m (Raj *et al.*, 2021). They are usually distributed in the hilly regions of Asian countries like India, China, Nepal, Bhutan, Japan, Pakistan etc. Around 79 species are present in the Western and South Western regions of China (Kshirsagar *et al.*, 2019). According to Joshi and Joshi, 2008, 29 species of *Swertia* are distributed throughout the mountainous regions of Nepal. Distribution ranges from 600 m (*S. angustifolia*) to 5600 m (*S. petiolata*). In India, approximately 40 species have been recorded till date, which are randomly distributed in the

western and eastern Himalayas. Various species of this genus are also found in the Western Ghats and Eastern Ghats in the state of Sikkim, Kerala, Maharashtra, Darjeeling, Arunachal Pradesh etc. In the North – Western Himalayas of India which includes states like Kashmir, Himanchal Pradesh, Uttarakhand are very rich in diversity of *Swertia*, specifically in the mountainous regions of the tropical area with an altitudinal range of 3600 – 13000 ft (Raj *et al.*, 2021). Some of the species that have been reported in Uttarakhand are *S. alternifolia*, *S. caroliniensis*, *S. chirayata* (Kaur *et al.*, 2019), *S. nervosa*, *S. angustifolia*, *S. alata* and *S. paniculata* (Gupta *et al.*, 2011) *S. cordata* (Bhandari *et al.*, 2006) etc.

The members representing the genus *Swertia* are annual, biennial or perennial herbs. Plant height ranges from 2 – 4 cm to 1.5m. The stems are absent, scapiform or well defined, erect and dichotomously branched from the base. The smallest plant of this genus is *S. acaulis* (few centimeters) found in the Himalayas while the largest is *S. scandens* (3 - 4m) found in Africa. The roots are fibrous or woody in some species with few secondary rootlets or rhizome with few fleshy adventitious rootlets. The leaves are opposite and rarely alternate or whorled or rosulate, sessile, sub- sessile or, petiolate and have entire margins. Some species possess narrow to long leaves, few species may have heart shaped leaves. The inflorescence is mainly cymose, usually grouped into simple or paniculate thyrses. The flowering and fruiting season were recorded from late August to October species and from September to December, respectively. The color of flowers varies greatly from species to species but most of species bear blue, purple, or greenish yellow flowers. The flowers are pedicellate, 4 – 5 merous. Calyx and corolla are rotate, lobed to base, tube less than 3 mm. The nectaries are 1 – 2 in number per corolla lobe, which are naked or covered by scales or flaps. The nectaries are glabrous, fringed or fimbriated in appearance. The number of stamens is similar to number of corolla lobes, attached at to base of corolla lobe sinuses. Stamens are sometimes surrounded by long hairs. Style may be short or elongated. The stigma is bilobed. Fruit is a capsule, that is enclosed by a persistent calyx and corolla that dehiscence from the top during seed dispersal. The seed are very small (Rijal, 2009).

The phytochemicals present in plant include several chemical constituents such as flavonoids, terpenoids, alkaloids, steroids, saponins etc. The medicinal importance any plant depends on the quality and quantity of these chemical constituents present in them. The genus *Swertia*, is blessed with a long list of phytochemical such as xanthenes, flavonoids,

seco-iridiods, iridiods, triterpenoids, alkaloids, and other secondary metabolites (Khanal *et al.*, 2015; Pan *et al.*, 2016). Li *et al.*, (2017) reported a list of flavonoids found in the genus *Swertia*, out of which xanthones have occupied the half of the total list. Xanthones are non-toxic which are used by plants for defense mechanism, due to which these herbs have wide range of uses, such as anti-tumor, anti-viral, anti-oxidant, immune modulatory and neuro-protective effects (Boshra and El-Haddad, 2018). Few triterpenoids and steroids have also been reported by different researcher from different species of *Swertia*. The demand of these components has increased due to their wide range of uses (Kshirsagar *et al.*, 2015). And in the past few decades a lot of efforts have been made so as to isolate and quantify these components, and various technologies and methods are being developed to maximize the yield of these components. The other two important bio active compounds have been reported from the genus *Swertia* are amarogrtin and swertiamarin. They have been isolated from different *Swertia* species. In addition to these, mangiferin compounds have also been isolated by different researchers. Kshirsagar *et al.*, 2016 isolated mangiferin from 11 species of *Swertia* and the results showed that the highest amount of mangiferin was present in *S. minor* after *S. chirata* compared to the studied *Swertia* species. Some other metabolites such as amines, lignins, naphthoquinones, monosaccharide, benzoates, phenols and aliphatic compounds, have also been reported in a report by Li *et al.*, 2017. As *Swertia* is the most potent medicinal herb and is critically endangered in most part of the world, screening out the potential source and best substitute for this plant is a compelling need.

The exploration for natural sources of antioxidant have tremendously increased since last few decades as they reduce the oxidative damage of the cell that is caused due to the production of free radicals (ROS and RNS) from variety of biochemical reactions taking place in the cell. Normally, antioxidants occurs in humans to keep a balance between oxidation and anti-oxidation, but due to exposure of smoking, alcohol, radiation, stress or environmental toxins induces the production of excessive ROS and RNS, which disrupt the balance between oxidation and antioxidation resulting in some chronic and degenerative diseases. In the context of *Swertia*, many researchers have evaluated the free radical scavenging activity of different species of *Swertia* through various assays. Chen *et al.*, 2011 evaluated the in vitro and in vivo antioxidant effects of methanolic and ethanolic extracts of *S. chirata* using different assays in mice. The results indicated strong in vitro antioxidant

activities of ethanolic extracts of *S. chirata*. Nag *et al.*, 2015 compared the antioxidant property of leafy shoots of three species of *Swertia* (*S. chirata*, *S. bimaculata*, and *S. decussate*) using DPPH, superoxide and nitric oxide radical scavenging activities and metal chelating properties. They observed that *S. chirata* had the best antioxidant potential among the three species. Srivastava *et al.*, 2018 did a comparative evaluation of iridoid and flavanoids present in 7 different species of *Swertia* present in India. They tested the antioxidant potential of *S. nervosa*, *S. bimacula*, *S. chirata*, *S. ciliata*, *S. paniculata*, *S. cordata*, and *S. alata* as well as of six phytochemicals compounds viz. (Swertiamarin, Sweroside, Swertisin, Amarogentin, Luteolin and Genistein). From the findings, it was observed that *S. chirata* had the highest antioxidant potential and had highest percentage of swertiamarin and other compounds as compared to other species that were used in the study. The other documented reports on the antioxidant potential of *Swertia spp.* indicates moderate to good antioxidant potential against different free radicals.

Since the past immemorial, the plants and their products have been used to preserve food in the form of spices, oils etc, as they have the ability to stop microbial decay. *Swertia* has been used from several years in various medicinal practices for its anti microbial activities. Scientists have reported anti microbial activity in various species of *Swertia*. Laxmi *et al.*, 2011 investigated the anti microbial activity of methanolic and aqueous extract *S. chirata* against 10 bacteria and 3 fungi using agar diffusion method. The activity was measured in term of diameter of zone of inhibition in millimeters (mm). It was observed that both the extracts showed concentration dependent activity. Bader, 2014 studied the anti microbial activity of extract of rhizome of *S. petiolata* against 4 human pathogenic bacteria. The results showed that hydro methanol extract showed excellent anti microbial properties against Gram- positive bacteria (*Staphylococcus aureus* and *Bacillus subtilis*), Gram-negative bacteria (*Escherichia coli*). However chloroform and petroleum ether extracts did not show significant antibacterial activity in both the type of bacteria. Mahendran *et al.*, 2015 tested the anti microbial activity of different compounds isolated from *S. corymbosa* against six human pathogens and found out that 1,2-dihydroxy-6-methoxyxanthone-8-O-d-xylopyranosyl followed by 1,8-dihydroxy-2,6-dimethoxyxanthone. Other than the anti bacterial activity various researchers have also evaluated anti fungal activity on some of the species of the *Swertia* (Afzal *et al.*, 2021).

Inflammation in the body is considered as the primary physiological defense mechanism that helps our body to protect itself against infections, burns, allergies etc. A persistent inflammation might act as an etiologic factor for many other chronic diseases. Plant materials have been used since ages to reduce inflammation and *Swertia* is one of them. It exhibits excellent anti inflammatory properties and has been used in herbal medicines for the same purpose. Banerjee *et al.*, 2000 studied the anti inflammatory effects of the xanthones derived from *S. chirata* in acute, sub-acute and chronic experimental models in male albino rats. The results indicated that 1,5-dihydroxy-3,8-dimethoxy xanthone had significant antiinflammatory action in all types of inflammations. Hossain *et al.*, 2012 investigated the *in vitro* anti- inflammatory activity of ethanolic extract of *S. chirata* using albumin denaturation. The results showed mean inhibition of protein denaturation 45.31 ± 0.000576 whereas, for control group it was found to be 50.00 ± 0.00177 . Bader *et al.*, 2017 made a study on the anti inflammatory property of aqueous and hydro- alcoholic extract of the rhizome of *S. petiolata* against carrageenan induced rat hind paw edema. The result of Hydro- alcoholic extract were seen to be better than the aqueous extract as the hydro- alcoholic extract decreased the inflammation by 66.66%, 68.10% and 77.7% at 1, 2 and 3 hours post treatment, while as the aqueous extract decreased the inflammation by 68%, 68% and 70.30% respectively.

4. Materials and Methods

The survey and the collection of the plants will be done from different parts of Uttarakhand Himalaya followed by the identification and authentication from BSI, Dehradun. Exploration of valley of flowers, Harshil, Gangotri and Chakrata has been done from 8 September to 20 November 2021. The samples will be air dried and later pulverized. For the extraction, 50gm of pulverized sample will be subjected to Soxhlet extractor with different solvent (Petroleum Ether, Ethyl acetate, Chloroform, Ethanol, and Water). The process of extraction in a Soxhlet extractor will be done for 16-17 cycle continuously. After complete extraction extract was concentrated.

1) Qualitative phytochemical analysis

Phytochemical screening:

The qualitative chemical tests of various extract of plant extract will be carried out by using standard procedure to determine the presence of various phytochemicals. The phytochemical screening will be done for the presence of glycosides, flavonoids, phenols, resin and tannins (Harborne, 1984; Kokate *et al.*, 2001).

1. Tests for carbohydrates

Molisch test: To 2 ml of the extract, add 1 ml of α -naphthol solution, and concentrated sulphuric acid through the sides of test tube. Purple or reddish violet colour at the junction of the two liquids reveals the presence of carbohydrates.

Fehling's test: To 1 ml of the extract, add equal quantities of Fehling's solution A and B, upon heating formation of a brick red precipitate indicates the presence of carbohydrates.

Benedict's test: To 5 ml of Benedict's reagent, add 1 ml of extract solution and boil for 2 minutes and cool. Formation of a red precipitate shows the presence of carbohydrates.

2. Tests for alkaloids

Dragendorffs test: To 1 ml of the extract, add 1 ml Dragendorffs reagent, an orange red precipitate indicates the presence of alkaloids.

Wagner's test: To 1 ml of the extract, add 2 ml of Wagner's reagent, the formation of a reddish brown precipitate indicates the presence of alkaloids.

Mayer's test: To 1 ml of the extract, add 2 ml of Mayer's reagent, a dull white precipitate reveals the presence of alkaloids.

Hager's test: To 1 ml of the extract, add 3 ml of Hager's reagent, the formation of yellow precipitate confirms the presence of alkaloids.

3. Tests for proteins and amino acids

Biuret test: To 1ml of the extract add 1ml of 40% sodium hydroxide solution and 2 drops of 1 % copper sulphate solution. Formation of violet colour indicates the presence of proteins.

Xanthoprotein test: To 1ml of the extract add 1 ml of concentrated nitric acid. A white precipitate is formed, it is boiled and cooled. Then, 20% of sodium hydroxide or ammonia is added. Orange colour indicates the presence of aromatic amino acids.

Lead Acetate test: To the extract, 1 ml of lead acetate solution is added. Formation of a white precipitate indicates the presence of proteins.

Ninhydrin test: Add two drops of freshly prepared 0.2% ninhydrin reagent to the extract solution and heat. Development of blue colour reveals the presence of proteins, peptides or amino acids.

4. Tests for tannins and phenolics

- i. To 1 ml of the extract, add ferric chloride, formation of a dark blue or greenish black colour product shows the presence of tannins.
- ii. To the extract, add potassium dichromate solution, formation of a precipitate shows the presence of tannins and phenolics.

5. Test for flavonoids

Shinoda Test: To 1 ml of the extract, add magnesium turnings and 1-2 drops of concentrated hydrochloric acid formation of red colour shows the presence of flavonoids.

6. Test for triterpenoids

Dissolve two or three granules of tin metal in 2 ml thionyl chloride solution. Then, add 1 ml of the extract into the test tube. The formation of a pink colour indicates the presence of triterpenoids.

7. Tests for steroids

Libermann Burchard test: Dissolve the extract in 2 ml of chloroform in a dry test tube. Add 10 drops of acetic anhydride and 2 drops of concentrated sulphuric acid. The solution becomes red, then blue and finally bluish green, indicates the presence of steroids.

Salkowski test: Dissolve the extract in chloroform and add equal volumes of concentrated sulphuric acid. Formation of bluish red to cherry red colour in chloroform layer and green fluorescence in the acid layer represents the steroid components in the tested extract.

Liebermann's reaction: Mix 3ml – extract with 3ml acetic anhydride, heat and cool, add few drops of concentrated sulphuric acid, blue color appears.

8. Test for saponins

About 1 ml of extract is diluted separately with distilled water to 20 ml, and shaken in a graduated cylinder for 15 minutes. A 1cm layer of foam indicates the presence of saponins.

9. Tests for fixed oils

Spot test: Press a small quantity of extract between two filter papers. Oil stains on paper indicate the presence of fixed oils.

Saponification test: To 1 ml of the extract add few drops of 0.5N alcoholic potassium hydroxide along with a drop of phenolphthalein. Heat the mixture on a water bath for 1-2 hours. The formation of soap or partial neutralization indicates the presence of fixed oils.

10. Tests for glycosides

Legal test: Dissolve the extract in pyridine and add sodium nitroprusside solution to make it alkaline. The formation of pink red to red colour shows the presence of glycosides.

Baljet test: To 1 ml of the test extract add 1 ml sodium picrate solution and the yellow to orange colour reveals the presence of glycosides.

Borntrager's test: Add a few ml of dilute sulphuric acid to 1ml of the extract solution. Boil, filter and extract the filtrate with chloroform. The chloroform layer is treated with 1 ml of ammonia. The formation of red colour shows the presence of anthraquinone glycosides.

Keller Kiliani test: Dissolve the extract in acetic acid containing traces of ferric chloride and transfer to a test tube containing sulphuric acid. At the junction, formation of a reddish brown colour, which gradually becomes blue, confirms the presence of glycosides.

11. Test for gums

Hydrolyse the test solution using dilute HCl. Perform Fehling's or Benedict's test. Red colour is developed.

12. Test for mucilages

- i. Powdered drug material shows red colour with ruthenium red.
- ii. Powdered drug swells in water

13. Test for Acidic compounds

- i. To the alcoholic extract sodium bicarbonate solution was added and observed for the production of effervescence.
- ii. A small amount of alcoholic extract was taken in warm water and filtered. The filtrate was then tested with litmus paper and methyl orange and observed for the appearance of blue colour.

2) Loss on Drying

This parameter is used to determine the amount of moisture present in a particular sample. The powder drug (10 g) sample will be placed on a tarred evaporating dish. The tarred evaporating dish was dried at 105⁰C for 6 h and weighed. The drying was continued until two successive reading matched each other or the difference between two successive weighing was not more than 0.25% of the constant weight.

3) Solvent extractive value:

The extraction of any crude drug with a particular solvent yields a solution containing different phyto-constituents. The composition of these phyto-constituents in that particular solvent depends upon the nature of the drug & solvent used.

4) Physico chemical constants:

Physico chemical parameters such as ash value, extractive value were performed according to the official method prescribed and the WHO guidelines on the quality control methods for medicinal plant material (Sati *et al.*, 2011).

5) Assay of Antioxidant Activity:

The ABTS, FRAP (Robu *et al.*, 2012; Benzie and Strain, 1996) and DPPH (Curcic *et al.*, 2012) Methods will be used for determination of antioxidant potential of the collected species of *Swertia*.

6) Antimicrobial Potential of Plant material:

Strains of different microbes will be chosen and Agar well diffusion method will be used to determine the antimicrobial activity of the collected specimens (Perez *et al.*, 1990).

7) Anti-inflammatory Activity:

For investigation of Anti-inflammatory potential of the collected specimens will be evaluated with Rat paw edema method (Winter *et al.*, 1962).

8) In vitro anti-inflammatory activity:

A mixture of lipoxygenase solution (20 µl), sodium phosphate buffer (160 µl, 0.1 mM, pH 7.0), along with sample extract (10 ml) will be incubated at 25°C for 5 min and the reaction is initiated using linoleic acid (10 µl) substrate solution. The formation of '(9Z, 11E)-13S)-13-hydroperoxyoctadeca-9, 11-dienoate' will be monitored by observing changes in absorption. For in vitro lipoxygenase inhibition assay, baicalein is used as a standard. IC₅₀ values will be calculated using the 'EZ-Fit Enzyme Kinetics program' (Lapchak *et al.*, 2007).

Future Prospects

The genus *Swertia* is being used for its medicinal properties since ages. The most commonly used species of this genus is *S. chirata*, also the most exploited one making it a critically endangered species in Indian Himalayan region and vulnerable in other parts of the world. The other species of this genus have not been explored much on their medicinal aspects hence a thorough comparative study of the various species from this genus could help in finding better alternative or a suitable substitute of *S. chirata* which in turn would help in its conservation. Also, not much work has been reported on other species of this genus in the chosen area of study and as the demand of medicinally important herbs is on its peak, this study will be of great use for the welfare of mankind.

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